

CXR-CLIP: Toward Large Scale Chest X-ray Language-Image Pre-training

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CXR-CLIP: Toward Large Scale Chest X-ray Language-Image Pre-training

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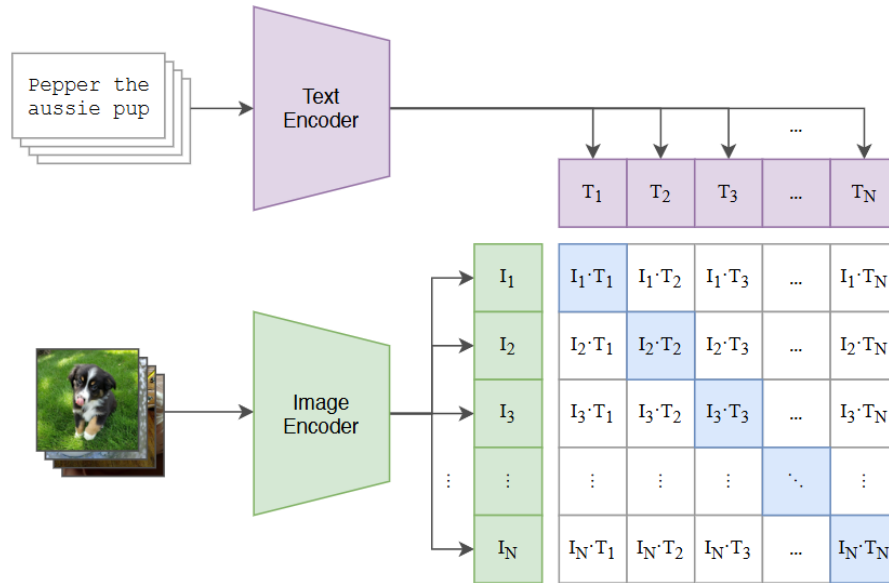


Medical Image Computing and
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MICCAI 2023
(MICCAI 2023)

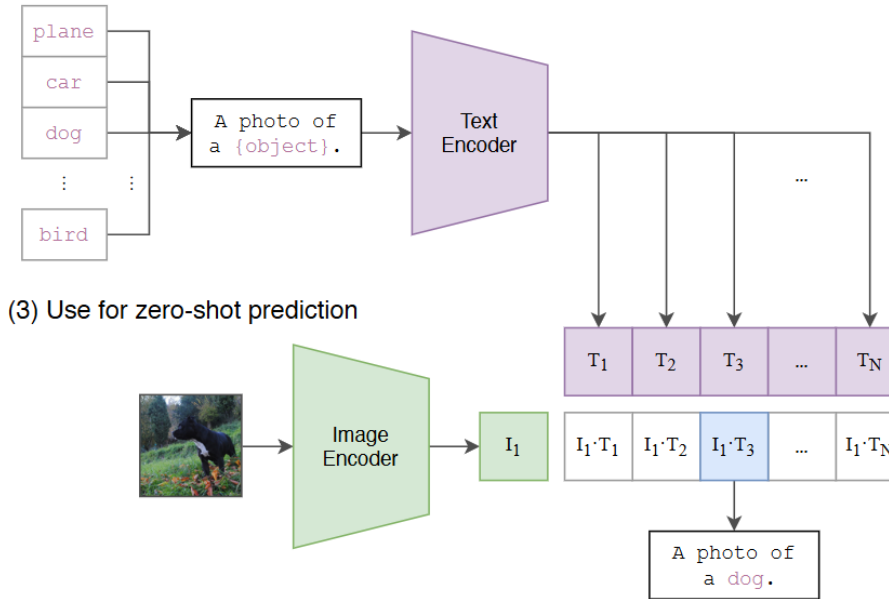
- What is CLIP?
- Introduction
- Method
- Experiment
- Conclusion

What is CLIP?

(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

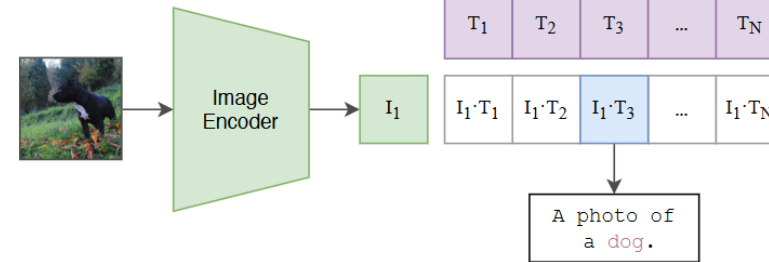


Figure 1. Summary of our approach. While standard image models jointly train an image feature extractor and a linear classifier to predict some label, CLIP jointly trains an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples. At test time the learned text encoder synthesizes a zero-shot linear classifier by embedding the names or descriptions of the target dataset's classes.



Learning Transferable Visual Models From **Natural Language Supervision**

ICML 2021, over **60k citations!**

What is CLIP?

$$L_{CLIP}(U, V) = -\frac{1}{2n} \left(\underbrace{\sum_{u_i \in U} \log \frac{\exp(v_i^\top u_i / \tau)}{\sum_{v_j \in V} \exp(u_i^\top v_j / \tau)}}_{\text{text} \rightarrow \text{image}} + \underbrace{\sum_{v_i \in V} \log \frac{\exp(u_i^\top v_i / \tau)}{\sum_{u_j \in U} \exp(v_i^\top u_j / \tau)}}_{\text{image} \rightarrow \text{text}} \right)$$

$v_i = \frac{f^i(E^i(x_i))}{\|f^i(E^i(x_i))\|}$: image encoder E^i 의 출력을 projection f^i 에 통과시킨 뒤 L2-normalize한 image embedding.

$u_i = \frac{f^t(E^t(t_i))}{\|f^t(E^t(t_i))\|}$: text encoder E^t 의 출력을 projection f^t 에 통과시킨 뒤 L2-normalize한 text embedding.

$v_i^\top u_j$: 두 unit vector 간 내적 = cosine similarity.

τ : learnable temperature scalar. logit의 scale을 조절하며, 값이 작을수록 softmax 분포가 sharp해짐

같은 image-text pair는 가까이 다른 image-text pair는 멀리 가도록 만들어줌!

- Chest X-ray을 학습을 시키고 싶은데 **고품질 임상 라벨을 구하기 어려운 문제**
- Vision-Language Pre-training (VLP) 이 라벨 없이 학습을 잘하더라 ...
 - CLIP, ALIGN
 - CXR로 학습을 시키자니 public 데이터셋은 저마다 **1) 다른 클래스로 데이터셋이 구성**이 되어 있고 그래서 **2) 데이터가 너무 적다.**
- Med-CLIP은 image-text랑 image-label 둘 다 썼던데, rule-based labeler의 성능에 의존하는 단점
 - 방사선 전문의가 만든 클래스별 프롬프트를 사용하여 이미지-텍스트 데이터와 이미지-레이블 데이터를 통합하는 학습 방법을 이용
- DeCLIP의 Multi View Supervision (MVS)를 변형해서 사용해보자!

$$(x^1, x^2, t^1, t^2) = \begin{cases} (x^1, x^2, t_{\text{findings}}, t_{\text{impression}}), & |X| \geq 2, T = \{t_f, t_i\} \\ (x^1, A_i(x^1), t_{\text{findings}}, t_{\text{impression}}), & |X| = 1, T = \{t_f, t_i\} \\ (x^1, x^2, t^1, A_t(t^1)), & |X| \geq 2, |T| = 1 \\ (x^1, A_i(x^1), t^1, A_t(t^1)), & |X| = 1, |T| = 1 \\ (x^1, x^2, p_1, p_2), \quad p_1, p_2 \sim \text{Concat}(\{p \sim P(y)\}_{y \in Y}) & \text{image-label data} \end{cases}$$

Table 5. Default positive and negative templates for suggested prompts, as well as class-specific templates. E is expressions for each class, + means text concatenation, [.] means random selection from the given list, and () is blank text.

	Positive templates	Negative templates
Default	[{E}., There is {E}., {E} is [present, seen, noted]., the presence of {E} is [seen, noted].]	[[There is, ()] + [no {E}., no radiographic evidence for {E}., no [visible, definite, obvious, appreciable, evident] {E}., no [convincing, definite, ()] evidence of {E}., no convincing signs of {E}]., No {E} is [visible, present, noted].]
Edema Pneumonia	[Default Positive templates, Findings are + [suggesting, compatible with, suggestive of, representing] + {E}.]	
Cardiomegaly	[heart size, cardiac size, cardiac silhouette, cardiac shadow, cardiac contour] + [is, appears] + [enlarged, increased].	[heart size, cardiac size, cardiac silhouette, cardiac shadow, cardiac contour] + [is, appears] + [normal, within normal limits, unremarkable].
Enlarged Cardio- mediastinum	[[cardiomediastinal, mediastinal] silhouette, [cardiomediastinum, mediastinum], mediastinal contour] + [is, appears] + [enlarged, widened].	[[cardiomediastinal, mediastinal] silhouette, [cardiomediastinum, mediastinum], mediastinal contour] + [is, appears] + [normal, within normal limits, unremarkable].
No Finding	[the lungs, both lungs, the lung fields, both lung fields] + [are clear, appear clear].	-

- **Image Encoder:**

- ResNet-50 or Swin-Tiny
- Embedding: global average pooling

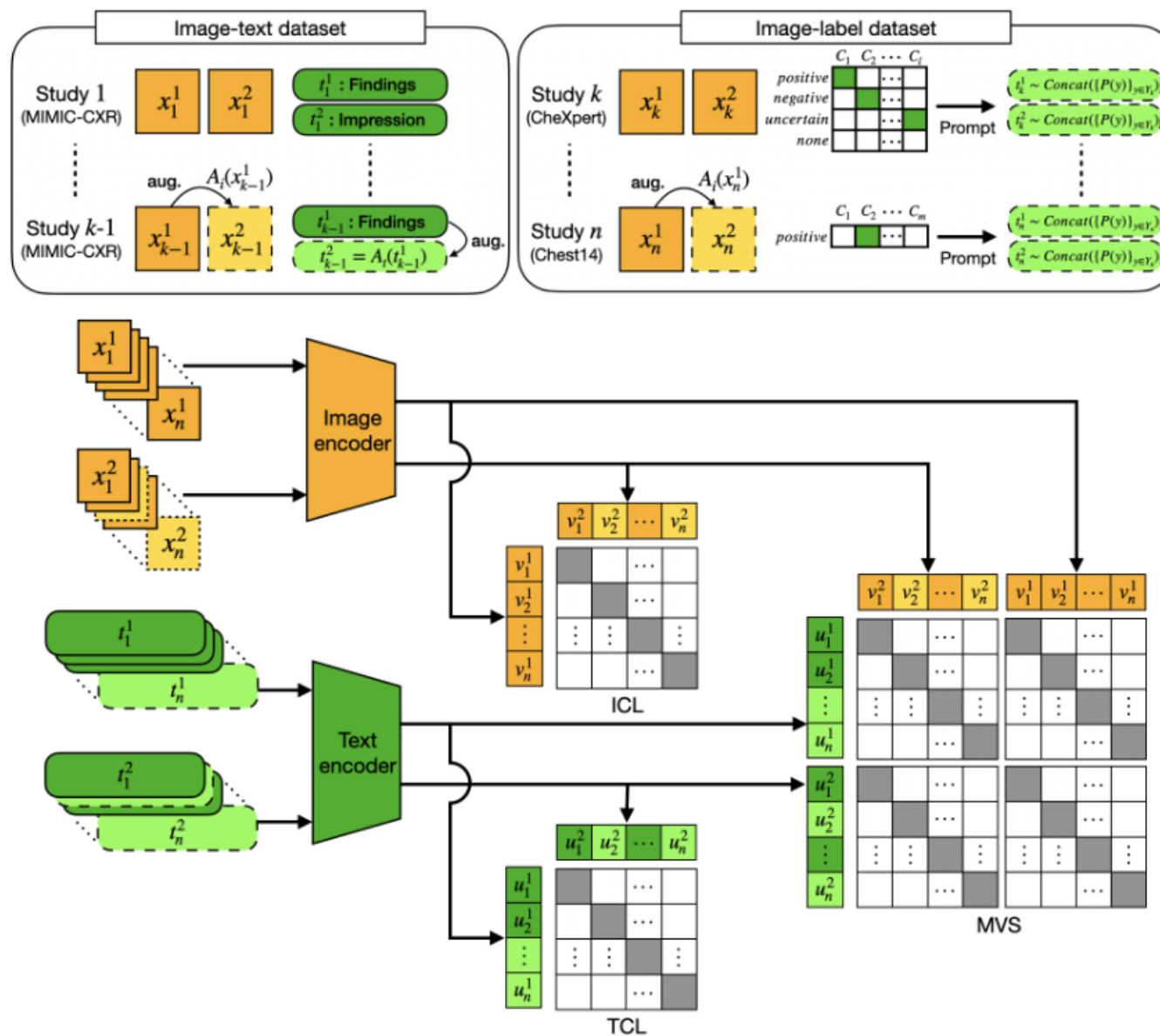
- **Text Encoder:**

- BioClinicalBERT - BERT와 architecture는 동일하지만, 의료 텍스트(MIMIC-III)로 pre-train된 버전
- Embedding: [EOS] token

- Embedding을 공통 차원으로 projection

$$V = \{v_i\}_{i=1}^n, U = \{u_i\}_{i=1}^n, (v_i, u_i) \text{ is paired}$$
$$v = \frac{f^i(E^i(x))}{\|f^i(E^i(x))\|}, u = \frac{f^t(E^t(t))}{\|f^t(E^t(t))\|}$$

Method - Loss Function



$$L_{CLIP}(U, V) = -\frac{1}{2n} \left(\underbrace{\sum_{u_i \in U} \log \frac{\exp(v_i^\top u_i / \tau)}{\sum_{v_j \in V} \exp(u_i^\top v_j / \tau)}}_{\text{text} \rightarrow \text{image}} + \underbrace{\sum_{v_i \in V} \log \frac{\exp(u_i^\top v_i / \tau)}{\sum_{u_j \in U} \exp(v_i^\top u_j / \tau)}}_{\text{image} \rightarrow \text{text}} \right)$$

$$L_{MVS} = \frac{1}{4} (L_{CLIP}(U^1, V^1) + L_{CLIP}(U^2, V^1) + L_{CLIP}(U^1, V^2) + L_{CLIP}(U^2, V^2))$$

$$L_{ICL} = L_{CLIP}(V^1, V^2)$$

$$L_{TCL} = L_{CLIP}(V^1, V^2)$$

$$L = L_{MVS} + \lambda_I L_{ICL} + \lambda_T L_{TCL}$$

Table 1. The number of studies for each dataset and split in this paper

Data Split	Pre-training			Evaluation			
	MIMIC-CXR	CheXpert	ChestX-ray14	VinDR	RSNA	SIIM	Open-I
Train	222,628	216,478	89,696	12,000	18,678	8,422	
Valid	1,808	233	22,423	3,000	4,003	1,808	
Test	3,264	1,000		3,000	4,003	1,807	3,788

Task	Dataset
1. Pre-training	MIMIC-CXR, CheXpert, ChestX-ray14
2. Retrieval	MIMIC-CXR, CheXpert5x200, Open-I
3. Classification	VinDR-CXR, RSNA, SIIM, CheXpert5x200

Augmentation 전략과 implementation details

구분	항목	설정
Image Augmentation (A_i)	Resize & Crop	Scale ([0.8, 1.1])
	CLAHE	적용
	Color Jittering	Random
	Brightness	([0.9, 1.1])
	Hue	
	Contrast	([0.8, 1.2])
	Saturation	
Text Augmentation (A_t)	Sentence Swap	적용
	Back-Translation	Italian → English

구분	항목	설정
Input / Embedding	Image Size	(224 × 224)
	Final Embedding Size	512
Loss Weight	(λ_I)	1.0
	(λ_T)	0.5
Training	Epochs	15
	Precision	Mixed Precision
	Early Stopping	Validation Loss 기준
	Optimizer	AdamW
	Initial Learning Rate	(5×10^{-5})
	Weight Decay	(1×10^{-4})
	LR Scheduler	Cosine Annealing
	Warm-up	1 epoch
Batch	Studies	128 studies
	Image-Text Pairs	256 pairs
Hardware / Framework	Framework	PyTorch
	GPU	4 × NVIDIA V100

Table 2. Comparison with state-of-the-art for zero-shot(ZS) or few-shot(10%) classification tasks. M, C, and C14 mean MIMIC-CXR, CheXpert, and ChestX-ray14, respectively. C* means CheXpert with reports, which are not publicly available. ResNet50 (R_{50}) and SwinTiny ($SwinT$) mean the image encoder used for each model.

Model Name	Pre-train Dataset	VinDR-CXR			RSNA			SIIM			C5x200 ZS-ACC
		ZS	10%	100%	ZS	10%	100%	ZS	10%	100%	
GloRIA R_{50}	C*	78.0	73.0	73.1	80.6	88.2	88.5	84.0	91.5	91.9	62.4*
CXR-CLIP R_{50}	M	78.8	82.1	82.2	83.3	88.5	89.2	85.2	88.3	90.5	56.2
CXR-CLIP $SwinT$	M	78.3	84.9	85.4	81.3	88.0	88.4	85.5	86.9	88.3	54.3
MedCLIP $SwinT$	M,C	82.4	84.9	85.1	81.9	88.9	89.0	89.0	90.4	90.8	59.2
CXR-CLIP R_{50}	M,C	83.0	81.4	82.1	81.7	88.5	88.9	86.4	88.4	90.7	61.7
CXR-CLIP $SwinT$	M,C	82.7	86.1	86.7	84.5	88.1	88.8	87.9	89.6	91.2	60.1
CXR-CLIP R_{50}	M,C,C14	78.1	80.2	81.0	81.8	88.7	89.3	85.2	91.5	92.8	60.3
CXR-CLIP $SwinT$	M,C,C14	78.9	88.0	89.0	80.1	89.2	89.8	91.4	92.9	94.0	62.8

Table 3. Comparison with state-of-the-arts for image-to-text retrieval. The notations of datasets and models are same to Table 2.

Model Name	Pre-Train Dataset	CheXpert5x200			MIMIC-CXR			Open-I			Total RSUM
		R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	
GloRIA R_{50}	C*	17.8	38.8	49.9	7.2	20.6	30.3	1.5	4.4	6.5	177.0
CXR-CLIP R_{50}	M	9.4	23.0	32.6	21.4	46.0	59.2	3.8	8.2	12.3	216.9
CXR-CLIP $SwinT$	M	8.4	21.5	30.2	21.6	48.9	60.2	3.6	8.3	11.5	214.2
MedCLIP $SwinT$	M,C	2.6	3.0	3.6	1.1	1.4	5.5	0.1	0.4	0.7	18.4
CXR-CLIP R_{50}	M,C	5.5	19.2	27.4	20.2	45.9	58.2	3.5	8.2	12.0	200.1
CXR-CLIP $SwinT$	M,C	8.5	23.0	31.6	19.6	44.2	57.1	3.1	8.3	11.6	207.0
CXR-CLIP R_{50}	M,C,C14	5.7	18.0	28.3	19.7	44.4	56.4	2.3	6.7	10.1	191.6
CXR-CLIP $SwinT$	M,C,C14	7.0	20.1	29.7	20.9	46.2	58.8	2.4	6.6	9.4	201.1

Table 3. Evaluation prompts for zero-shot classification. For VinDR-CXR and SIIM, we use simple prompt in Jang et al. [10], and we use prompt in BioVIL [2] for RSNA.

Dataset	Positive	Negative
VinDR-CXR, SIIM	{classname}	No {classname}
RSNA-pneumonia	Findings suggesting pneumonia.	No evidence of pneumonia.

Table 4. Ablations and comparison with CLIP [23] and DeCLIP [14]. Our augmentations effectively preserves clinical meaning than EDA. Our full methodology (CXR-CLIP) outperforms DeCLIP.

Method	CheXpert 5x200				MIMIC-CXR			Total RSUM
	ACC	R@1	R@5	R@10	R@1	R@5	R@10	
Vanila CLIP	58.9	4.4	14.4	22.6	17.3	41.2	52.6	152.5
+ Study Level Sampling	58.7	4.6	15.1	23.2	17.8	42.5	54.2	157.4
+ Augmentations	60.6	5.7	17.0	24.9	16.1	40.2	51.5	155.4
+ MVS	61.2	5.4	17.1	24.7	16.3	40.6	53.3	157.4
+ ICL	61.6	6.8	20.3	28.6	17.5	41.6	53.2	168.0
+ TCL (CXR-CLIP)	61.7	6.2	18.2	29.1	19.6	44.8	56.6	174.5
MVS of DeCLIP (EDA)	59.5	3.2	15.5	22.9	15.8	39.1	51.5	148.0
MVS of DeCLIP (Our aug)	59.4	6.0	17.0	24.4	15.1	38.8	51.8	153.1
DeCLIP (Our aug)	59.4	5.7	16.1	24.6	18.1	44.0	55.3	163.8

- Data Scaling

- Image-label 데이터셋(CheXpert, ChestX-ray14)을 prompt를 통해 image-text pair처럼 변환
- 하나의 study 내 multiple image·multiple report section을 활용해 학습 데이터를 확장

- Trade-off 발견

- Image-label 데이터를 추가할수록
 - **Classification**(zero-shot, few-shot 포함) 성능은 **향상**
 - 반면 **retrieval** 성능은 **저하**

- ICL/TCL

- 각 modality 내에서의 discriminating power를 강화하는 추가 loss
- image-text retrieval 성능을 **효과적으로 끌어올리는 역할**